The GUIDES Framework:
Enhancing Engineering Education with GenAI

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Outline

1. The Generative AI Disruption: Promise & Peril for Education
2. The **GUIDES** Framework for GenAI Era Engineering Education

GROUNDED
H**U**MANE
INTENTIONAL
DIALOGIC
EXPERENTIAL
SELF-DEVELOPMENT

1

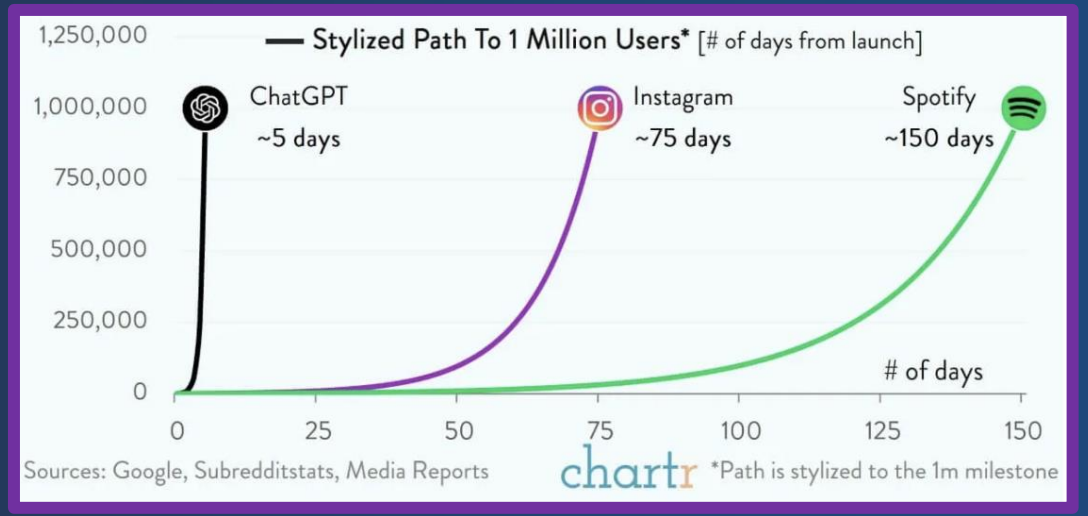
The Generative AI (GenAI) Disruption

Image Generator

Speech Synthesizer

Text Generator

Conversational AI (ChatBot)



The (disruptive) potential of GenAI as a Co-Pilot

CODE GENERATION

Give me code for the Dijkstra algorithm in Python

Here is the code for the Dijkstra algorithm in Python:

Copy code

```
from collections import defaultdict
import heapq

def dijkstra(graph, source):
  # Initialize distances and predecessor for each vertex
  distances = defaultdict(lambda: float('inf'))
  predecessor = defaultdict(lambda: None)
  distances[source] = 0

  # Initialize priority queue with all vertices and their distances from
the source
  priority_queue = [(0, source)]
  while priority_queue:
    # Extract vertex with minimum distance from the queue
    distance, vertex = heapq.heappop(priority_queue)

    # Update distances and predecessors for all neighbors of the vertex
    for neighbor, weight in graph[vertex]:
      if distances[neighbor] > distance + weight:
        distances[neighbor] = distance + weight
        predecessor[neighbor] = vertex
        heapq.heappush(priority_queue, (distances[neighbor], neighbor))

  return distances, predecessor
```

GitHub Copilot

DOI:10.1145/3570220

Matt Welsh

Viewpoint

The End of Programming

The end of classical computer science is coming, and most of us are dinosaurs waiting for the meteor to hit.



I CAME OF AGE in the 1980s, programming personal computers such as the Commodore VIC-20 and Apple][e at home. Going on to study computer science (CS) in college and ultimately getting a Ph.D. at Berkeley, the bulk of my professional training was rooted in what I will call "classical" CS: programming, algorithms, data structures, systems, programming languages. In Classical Computer Science, the ultimate goal is to reduce an idea to a program written by a human—source code in a language like Java or C++ or Python. Every idea in Classical CS—no matter how complex or sophisticated, from a database join algorithm to the mind-bogglingly obtuse Paxos consensus protocol—can be expressed as a human-readable, human-

GenAI going autopilot: implications for education

TECHNOLOGY *The Atlantic*

The College Essay Is Dead

Nobody is prepared for how AI will transform academia.

By Stephen Marche

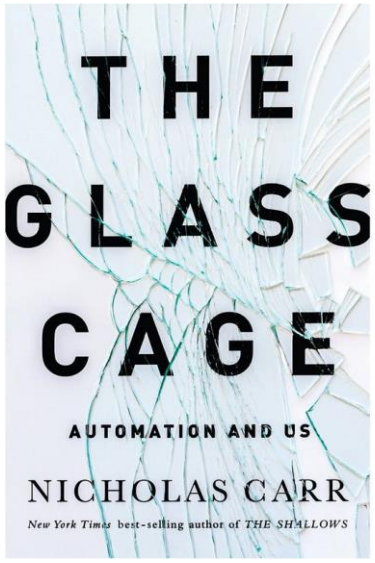
MOTHERBOARD
TECH BY VICE

ChatGPT Is Passing the Tests Required for Medical Licenses and Business Degrees

ChatGPT passes exams from law and business schools

By Samantha Murphy Kelly, CNN Business
Updated 1:35 PM EST, Thu January 26, 2023

ChatGPT passes Wharton Business School test: research paper





Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

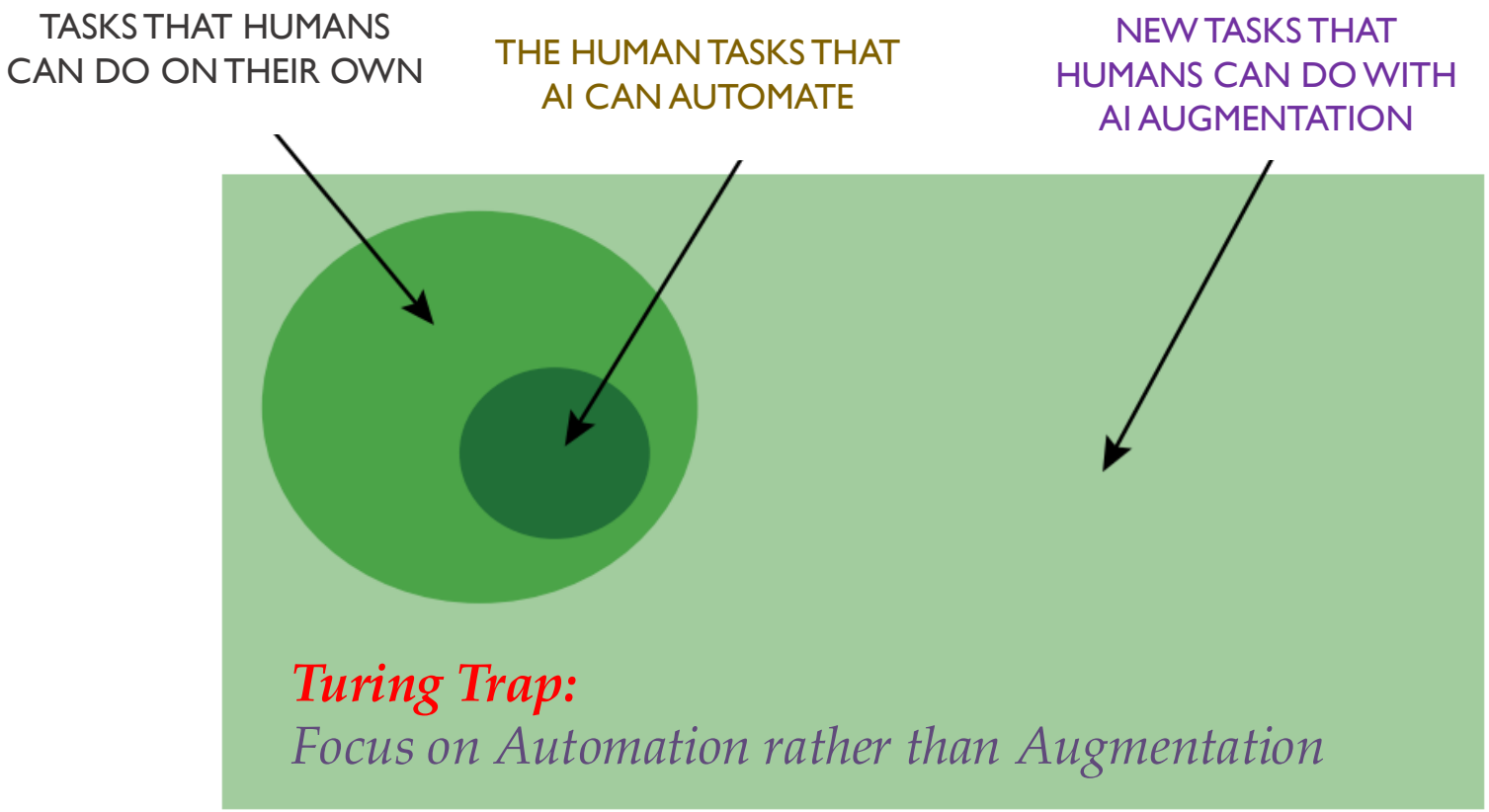
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IEEE EDUCON 2023
IEEE Global Engineering Education Conference

How automation creates its own set of challenges due to the automation complacency

The Human-Centered Nature of Education

Education is intrinsically human-centered: it is a human being we are educating for a purpose.

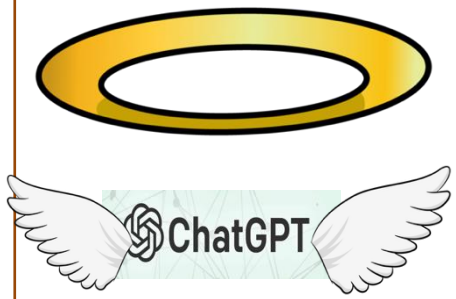


The role of technology is to support the education process and not to automate it

GenAI for Education: *Critical Appraisal of Potential Risks*

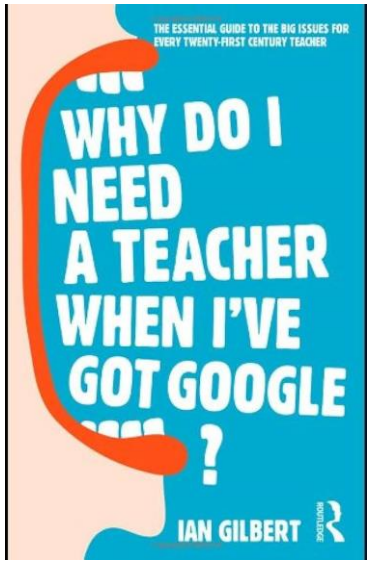
HALO EFFECT

The Gullibility to Believe AI is infallible



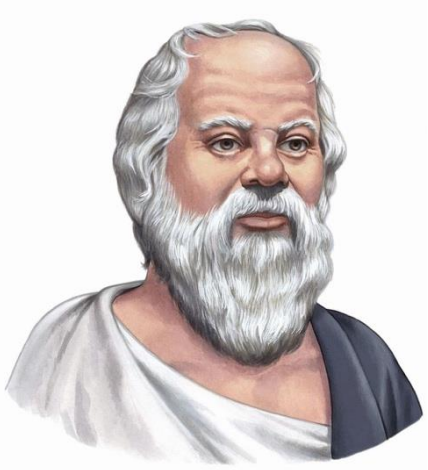
DISINCENTIVES

Why spend time and effort learning when you can just Google/ or use ChatGPT



SOPHISTRY & PRETENTION

Appearances become more important than truth. Easy to create plausible but untrue information.



TECHNOLOGICAL OUTSOURCING



AL-GHAZALI AND THE ROBBERS

HOMOGENIZATION

Making the thoughts/ outputs of every student ***formulaic*** *and cliched*



GenAI for Education: *Critical Appraisal of Potential Risks*

AI COLONIALISM

The "Global North"—the rich dominating countries stand to dominate and monopolize the market

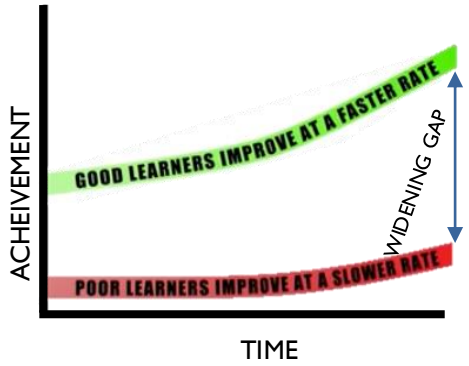


MARGINALIZING THE MARGINALIZED

Data-poor regions, not having indigenous LLMs, are put at a long-term risk of being colonized by standards embedded in LLMs

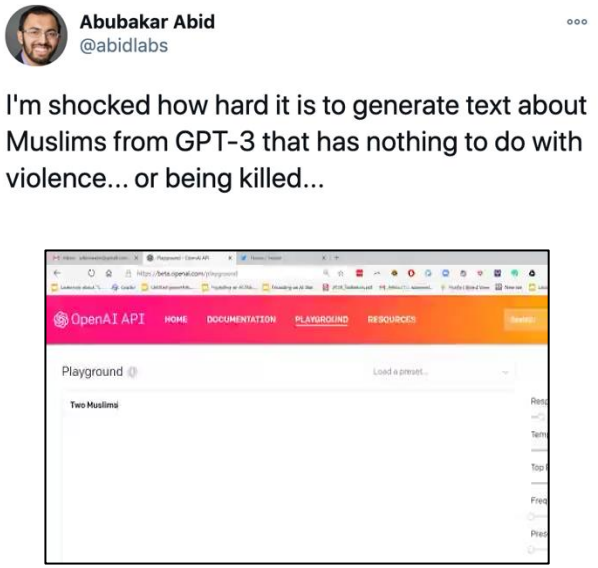
THE GENAI EDTECH MATTHEW EFFECT

Those who already know, stand to benefit most from GenAI, as they can prompt, critique and judge the output better



REINFORCING BIAS

Those who already know, stand to benefit most from GenAI, as they can prompt, critique and judge the output better



2

The GUIDES Framework

GROUNDED
H**U**MANE
INTENTIONAL
DIALOGIC
EXPERENTIAL
SELF-DEVELOPMENT

GUIDES

GROUNDED

To be able to use GenAI in Education, it is vital that it is reliable, trustworthy, and grounded in facts (i.e., it has strong foundations or is evidence-based)

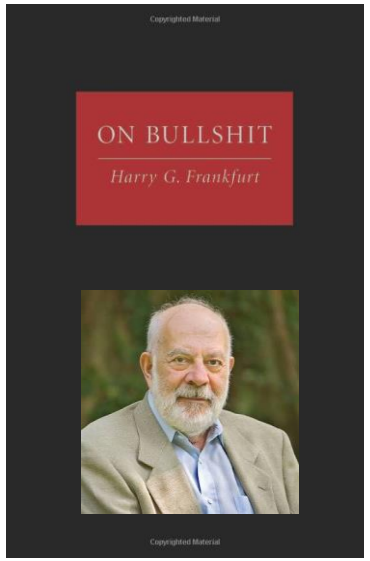
CAVEAT EMPTOR:

Modern AI models (some people call them foundation AI models) have been likened by some experts to:



CASTLES IN THE AIR

Hallucination and Confabulations



ChatGPT is a bullshit generator. But it can still be amazingly useful



Arvind Narayanan and Sayash Kapoor
Dec 6, 2022

38 10

The philosopher Harry Frankfurt defined bullshit as speech that is intended to persuade without regard for the truth.

Can we trust GenAI for critical tasks?

Let's assume that the GenAI becomes highly accurate (let's says 95% of the time)

Can we use it for critical tasks?
(if the working of the tool is opaque)



Looks can be deceiving

Find the derivative of the function f(x) = 3x^2 + 2x - 1. Describe your working

To find the derivative of the function f(x) = 3x^2 + 2x - 1, we need to use the rules of differentiation. The derivative of a function is a measure of how the function is changing at a particular point.

The derivative of a constant term (such as -1 in this case) is always 0, so we can ignore that part of the function. The derivative of a term with a coefficient of 1 (such as 2x in this case) is just the coefficient times the variable (in this case, 2x), so the derivative of 2x is 2.

The derivative of a term with a coefficient other than 1 (such as 3x^2 in this case) is the coefficient times the derivative of the term with the coefficient of 1 (in this case, 3 times x), so the derivative of 3x^2 is 3x.

Now that we have found the derivative of each term in the function, we can add them up to get the derivative of the whole function:

f'(x) = 3x + 2 = 6x + 2

This is the final answer. The derivative of the function f(x) = 3x^2 + 2x - 1 is f'(x) = 6x + 2.

Sam Altman @sama

ChatGPT is incredibly limited, but good enough at some things to create a misleading impression of greatness.

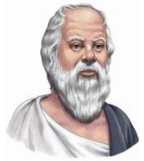
it's a mistake to be relying on it for anything important right now. it's a preview of progress; we have lots of work to do on robustness and truthfulness.

3:11 AM · Dec 11, 2022

Yann LeCun @ylecun

My unwavering opinion on current (auto-regressive) LLMs
1. They are useful as writing aids.
2. They are "reactive" & don't plan nor reason.
3. They make stuff up or retrieve stuff approximately.
4. That can be mitigated but not fixed by human feedback.
5. Better systems will come

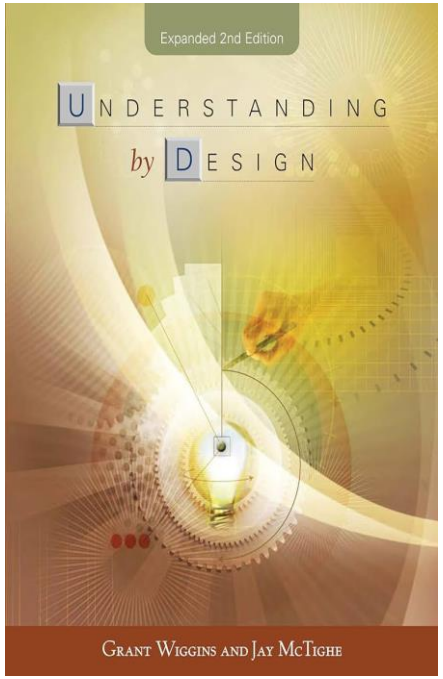
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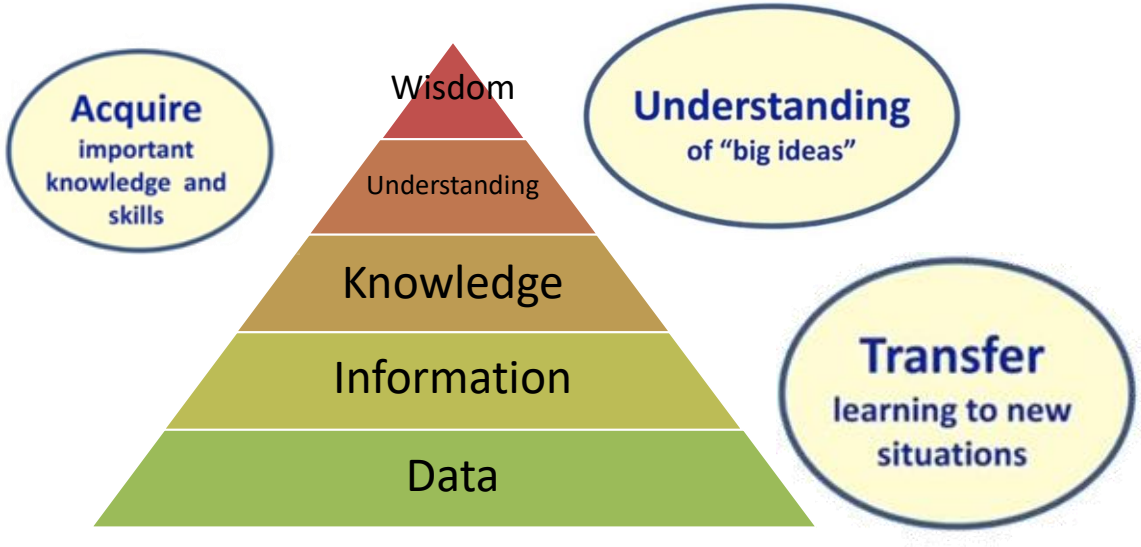
Socratic critique of technology

Such tools may impart "not truth but only the semblance of truth" and may appear to be omniscient and will generally know nothing."

Developing Critical Thinking/ Informed Judgement Skills



Indispensability of understanding/ transfer skills

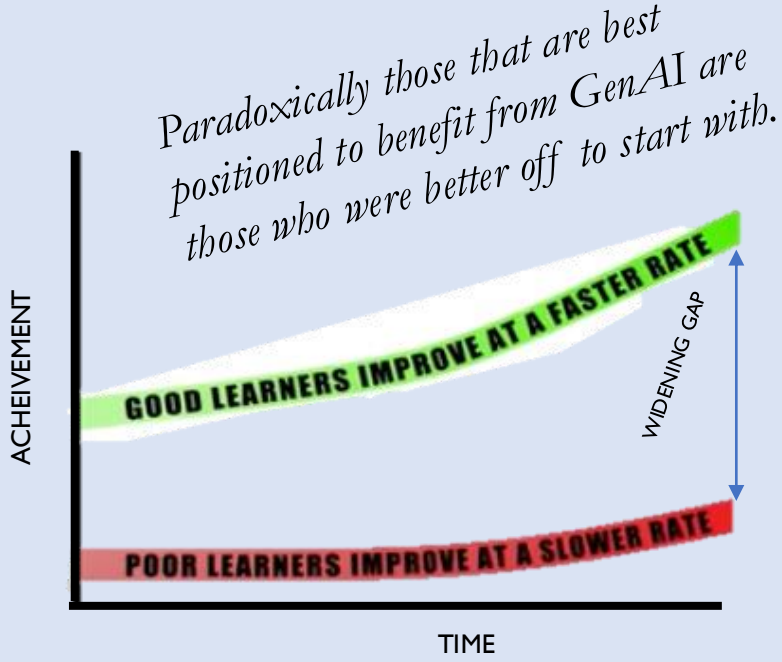


Cultivating Critical AI Literacy and Understanding Limitations

Questions/ Discussion Mini-Break

AI is not just ChatGPT. GenAI continues to evolve! Some ***technical approaches*** *for combating these deficiencies have been proposed (e.g., Retrieval Augmented Generation or RAG).*

Critical Thinking: *Can learning be automatized and outsourced? (in other words, can wisdom be automated)*



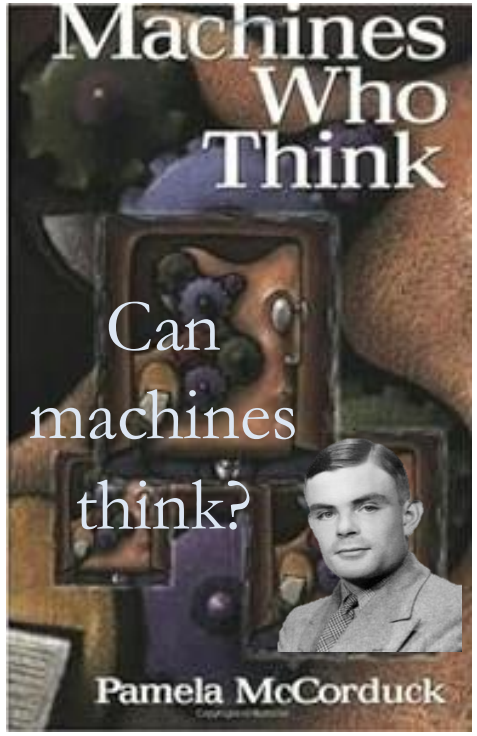
Technology can amplify human intent/ capabilities but NOT substitute for them.

GUIDES

HUMANE

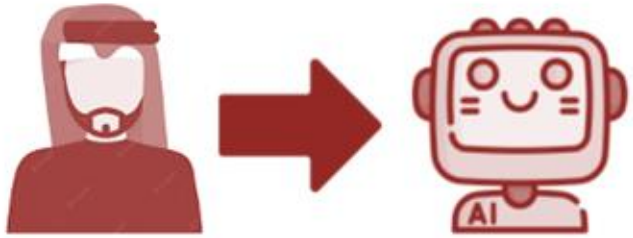
Education is a uniquely human process, emphasizing the complex social interactions and extended period of enculturation necessary to become an adult in human society.

Focus on machine rather than on human intelligence



Shifting Metaphors:
Machines like human brains
-> Our brains are like computers

Automation



Replace human intelligence with artificial intelligence.

Turing Trap:
Focus on Automation rather than on Augmentation

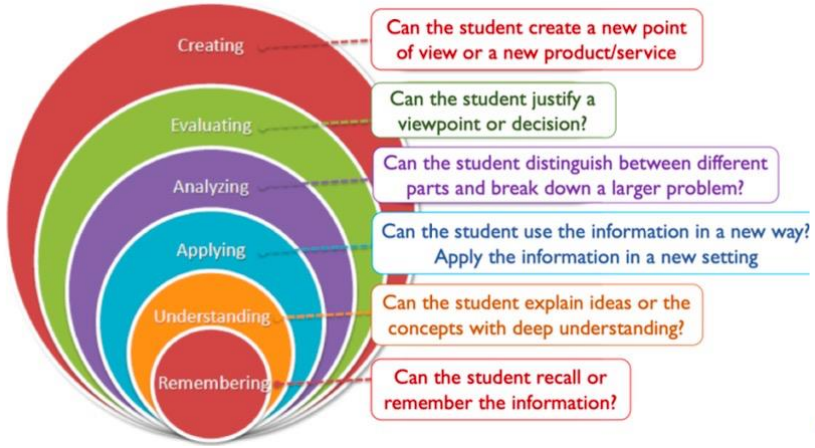
Augmentation



Augment human intelligence with artificial intelligence.

Human-Centered Focus:
Education needs to focus on humans and on how machines/ algorithms can support human learning

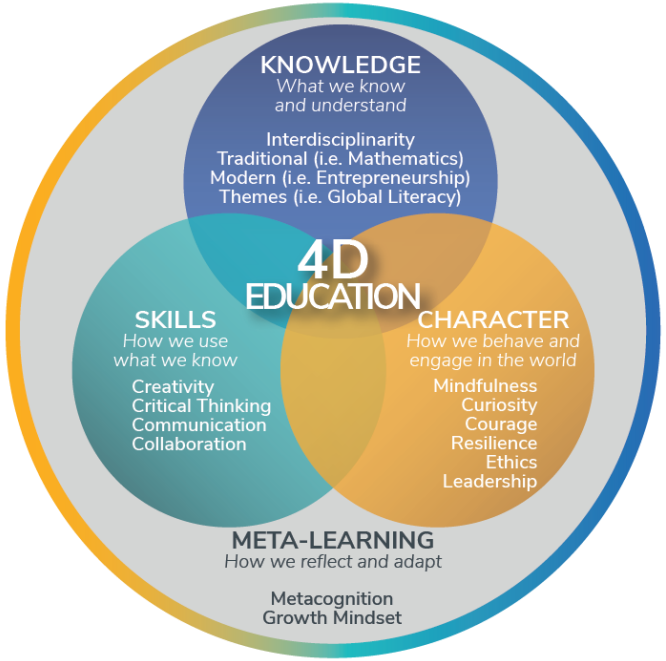
Holistic Learning: *Whole Person Education*



Revised Bloom's Taxonomy [Cognitive domain]
(Source: Krathwohl & Anderson)



Bloom's Taxonomy [Affective Domain]
(Source: Krathwohl, Bloom, and Masia)

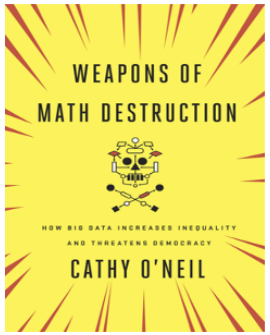


Artificial Intelligence is not very artificial..

"Whether or not it draws on new scientific research, technology is a branch of moral philosophy, not of science."— Paul Goodman

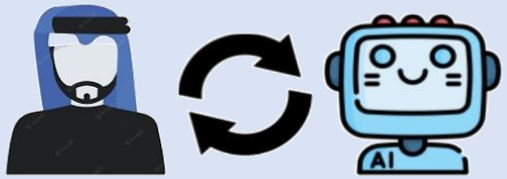
"

THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD

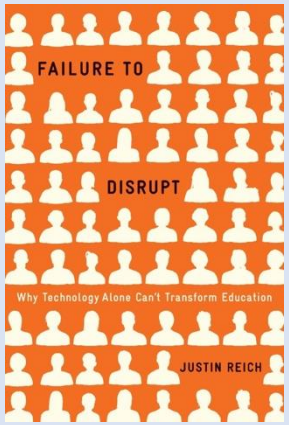


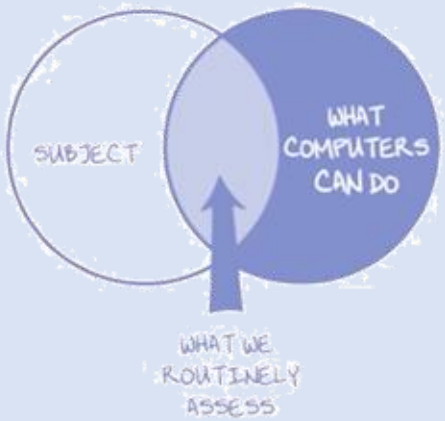
Education for ethical awareness as well as moral conduct

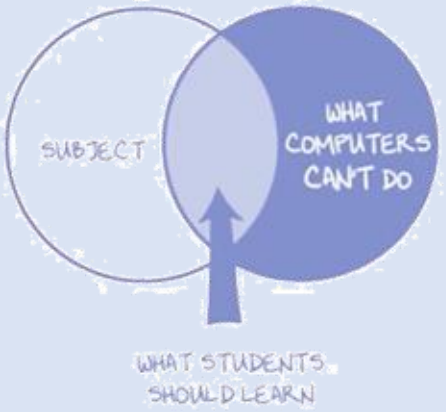
Questions/ Discussion Mini-Break

LIVING BEING  ENGINEERED ARTIFACTS (MACHINES/BOTS)

Emphasizing our unique human skills, which distinguish us from computers/ AI

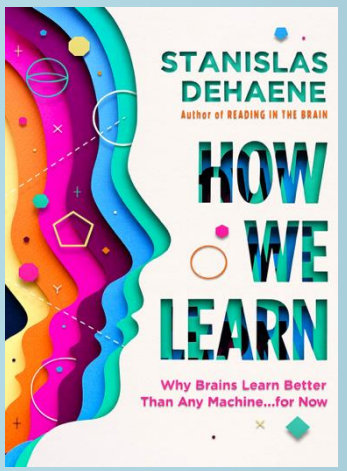






GUIDES

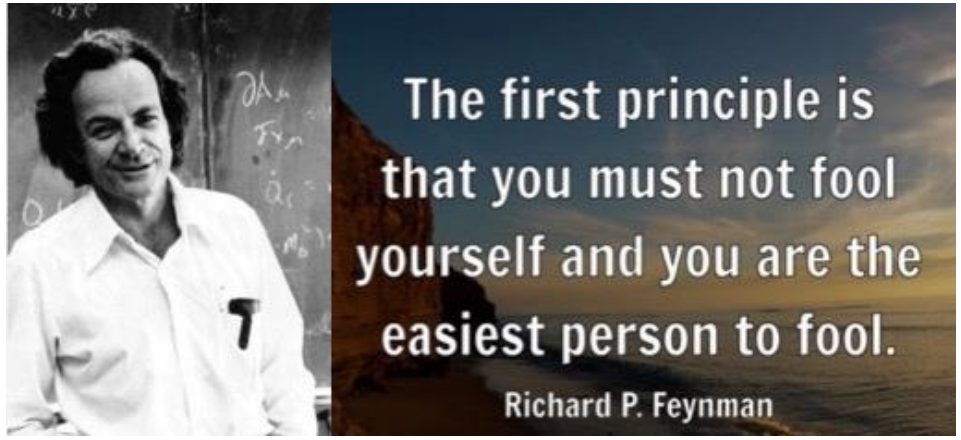
INTENTIONAL (*ACTIVE LEARNING*)

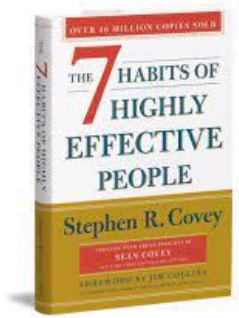


"Efficient learning means refusing passivity, engaging, exploring, and actively generating hypotheses and testing them on the outside world."

Law of the Harvest: Beware of Shortcuts

Time needed for meaning-making and Understanding:
(Otherwise, we could give all the answers/ info at the start)







Cramming, Outsourcing, Copying

Learning basics cannot be ignored when tech advances



Dollars Versus Diplomas

What every student should know: Seven learning impediments and their remedies

Junaid Qadir

Many well-intentioned students fail to thrive in a university environment, not due to lack of effort, but because of a deficient mind-set and/or methodology. While a wide variety of problems can result in suboptimal learning performance, there are seven main categories of learning impediments that hinder student learning with alarming regularity. You may be facing one, or more, of these impediments. Not to fret though; each of these debilitating impediments can be surmounted with self-awareness and some effort, paving the way for enjoyable, and productive, learning experiences. In this article, I will identify the offending learning impediments and propose plausible remedial policies to counter each.

Is there anyone so wise as to learn by the experience of others?

—Voltaire

Digital Object Identifier 10.1109/MPOT.2014.2368793
Date of publication: 6 May 2015

IMAGE LICENSED BY GRAPHIC STOCK

First impediment: Having a fixed mind-set

Psychologists have highlighted the self-enforcing propensity of the human mind-set in that human optimism and pessimism are both self-fulfilling. As Henry Ford said, "Whether you think you can or think you can't—you are right." A *fixed mind-set*, a mind-set model proposed by researcher Carol Dweck, takes the pessimistic approach to the concept of intelligence and skill development. It characterizes a belief that intelligence or talent are *fixed* human traits that cannot be significantly developed beyond a certain degree. Students with fixed mind-sets obsess over documenting their intelligence instead of focusing on developing their intelligence. Many fixed mind-set students are gifted students who have not learned how to struggle or the value of hard work. While they aced school subjects with little effort, when confronted with more significant intellectual challenges in a university setting that require more effort, these students capitulate easily while resorting to the excuse of a lack of aptitude. Why else would they—being such intelligent students—be confused?

What can be done: The antidote to this problem is to: 1) have a growth mind-set, 2) aim for mastery, and 3) use intrinsic motivation.

Catch phrase for the solutions: The sky is not the limit—your mind is. Conquer your mind to realize your potential.

Remedy 1: Have a growth mind-set

A *growth mind-set* student believes that hard work, grit, and

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Careers and Grad Ed

Learning 101: The untaught basics

Junaid Qadir and Muhammad Ali Imran

Despite the accessibility of a wealth of information in the current era—books, universities, and online massive open online courses (MOOCs)—well-intentioned and hard-working students often fail to learn effectively due to deficient learning techniques or improper mind-sets. Two things, in particular, hinder students from achieving their potential. First, the students' intuition regarding how learning works is often flawed and counterproductive; second, despite significant progress in the research discipline of "learning sciences," these hard-earned scientific insights have not yet filtered their way through the research community to the students who stand to benefit most from this knowledge.

In this article, we aim to popularize the important insights from learning-science researchers by making it accessible to students, who continue to largely use suboptimal intuitive learning techniques (not knowing any better). The benefits of becoming wise to these insights are immediate and substantial: we can use this knowledge to understand and calibrate our learning, and it can also facilitate efficient learning.

Digital Object Identifier 10.1109/MPOT.2017.2749408
Date of publication: 2 May 2018

Common learning impediments

To learn optimally, we need to discover how to learn effectively and also to identify and avoid common learning impediments. In our previous work, we identified seven common learning impediments (shown in Table 1). These impediments emerge from deficient mind-set (numbers 1 and 4), focus (2 and 3), support (5 and 6), and metacognition (7). These issues are not insurmountable; three potential remedies for each learning impediment are identified in our previous work as shown in Table 1. A major impediment—arguably the mother impediment of all the listed impediments, if considered broadly—is number 7, "Not learning how to learn." The overarching remedy for this problem is to spend time developing better metacognitive skills. Our aim is to succinctly present the relevant research findings in an accessible manner so that you can use this knowledge to improve your metacognitive skills and learning strategies.

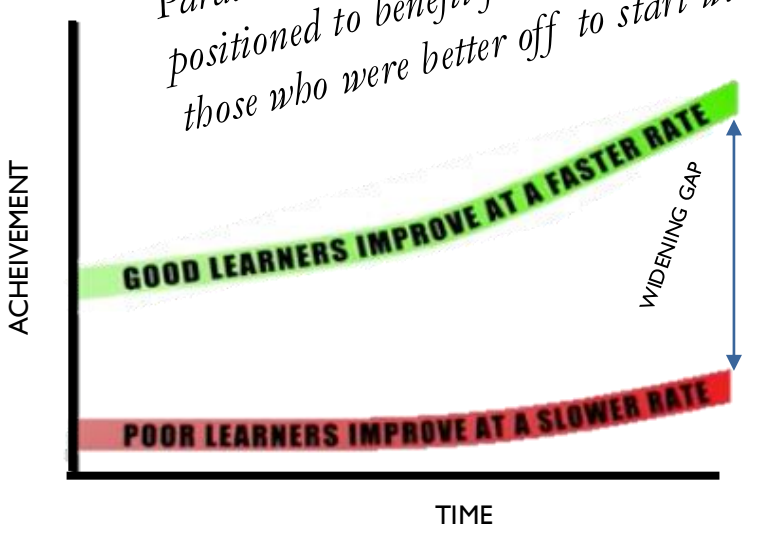
Learning how to learn

The most useful piece of learning for the uses of life is to unlearn what is untrue.

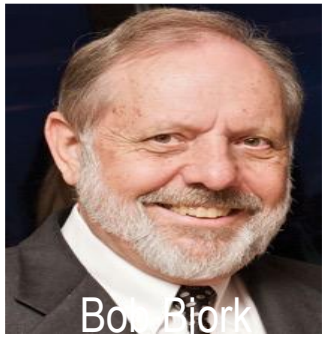
—Antisthenes

Despite our intimacy with the task of learning (through the trials and errors of everyday living), research has shown that our metacognitive judgment about learning (i.e., our intuitive understanding of how we learn) is flawed, and many of our well-established and commonly accepted intuitive notions are not only incognizant of scientific findings about learning but oppose the optimal course of action. Unfortunately, the common societal attitude, and the general assumption in colleges and universities, is that students can automatically figure out the art and science of learning by themselves. This is reflected in the way we emphasize the teaching of content and subjects in educational institutions but not the dissemination of information regarding an optimal learning strategy.

There is a Taoist saying that states, "To gain knowledge, one should add something every day, but to gain wisdom, one should subtract something every day." We propose that students should supplant the erroneous intuitive learning model with an improved metacognitive model of how learning



Paradoxes of Education: Importance of metacognition

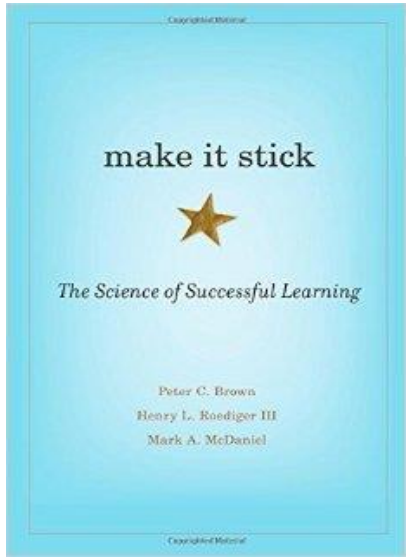


Theory of ***"Desirable Difficulties"***

It is beneficial for learning to create study conditions that make learning difficult and slow. This is in conflict with the quick-fix desire in learners and instructors to see quick improvements.

"One of the most striking research findings is the power of active retrieval—testing—to strengthen memory, and that **the more effortful the retrieval, the stronger the benefit**."

"by spacing or interleaving the practice, retrieval is harder, your performance is less accomplished, and **you feel let down, but your learning is deeper and you will retrieve it more easily in the future**."



Questions/ Discussion Mini-Break

Establishing the Baseline: Zero-Shot Learning

First Law 

When the AI is readily accessible, your work should be no worse than that produced by the AI alone.
(Embrace change and leverage AI)

Second Law 

(Corollary of Law 1)
When AI is there, and you don't want to be made irrelevant, maintain your competitive advantage
(Become **AI Proof**)

Third Law 

After becoming users of AI, don't let your inherent unaided capabilities drop
(**Don't outsource your thinking muscle allowing it to atrophy**; i.e., do no harm [*a la* Asimov; Hippocrates]

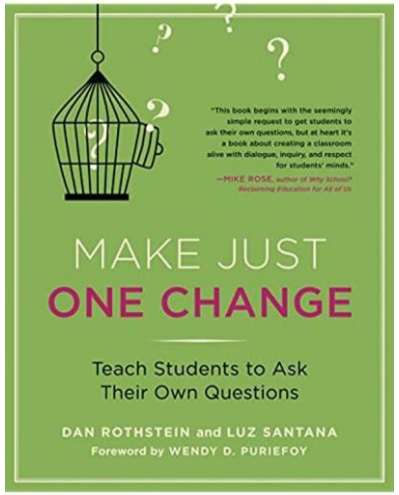
Beyond the Baseline: Enhancing Creativity and Critical Thinking

GUIDES

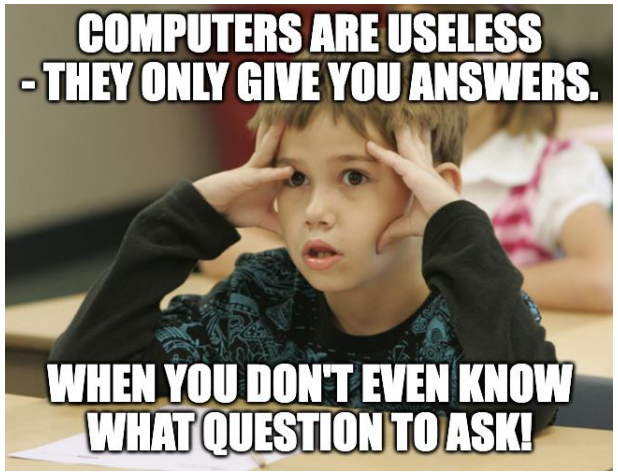
DIALOGIC

True education is less about the content covered in classes and more about what an individual discovers and creates with the resources available to them

Enhancing IQ (inquiry quotient)

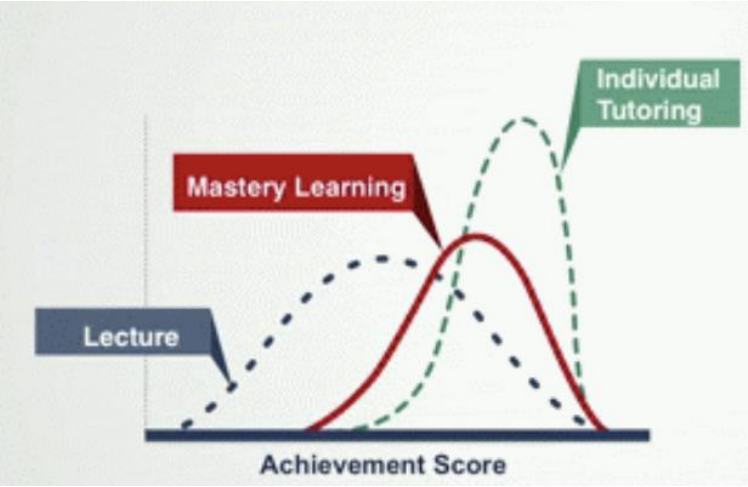


To engage students in meaning making and to develop self-directed learners,
students should be encouraged to develop their own questions—
and to pursue the answers through active inquiry (i.e., to become researchers)



Personalized Learning (Potential role as Virtual Tutor)

The Two-Sigma Effect of Individualized Tutoring



*Students learn best when they receive **formative feedback** about specific qualities of their work along with concrete advice on what they can do to improve.*

VIRTUAL TUTORING & PERSONALIZED LEARNING

JU

My teacher asked me, "What is the maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240?". My reply was 8 but my teacher said that this is wrong. Why is it wrong?

The maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240 is not 8. The subnet mask 255.255.255.240 allows for 4 bits to be used for host addresses, and each bit can represent 2 values (0 or 1). Therefore, the total number of host addresses that can be assigned in this subnet is 2^4 = 16.

It's important to note that the subnet mask is used to determine the number of host addresses that can be assigned in a subnet. The subnet mask consists of a series of 1s and 0s, and the number of 1s indicates the number of bits that are reserved for the network portion of the IP address. The number of 0s indicates the number of bits that are available for host addresses. By counting the number of 0s in the subnet mask and using the formula 2^n (where n is the number of 0s), you can determine the maximum number of host addresses that can be assigned in a subnet.

Educating for Human Intelligence Augmentation



Principle 1

The learning process should be centered around the student, rather than on the teacher, curriculum, or AI

(Be learner-centric and make your education human-centered)



Principle 2

AI should be utilized in a way that facilitates scalable iterative rapid ***formative feedback***

(Scale formative feedback using AI)

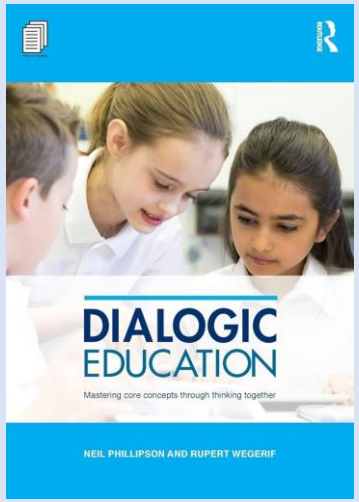


Principle 3

Teach the students the critical thinking skills necessary for judging and correcting the output of AI models

(Equip students with critical thinking to judge AI's output.]

Questions/ Discussion Mini-Break



Optimizing Learning through Few-Shot and Multi-Turn Interactions

Strategies for Integrating GenAI in Dialogic Activities

GUIDES

EXPERENTIAL

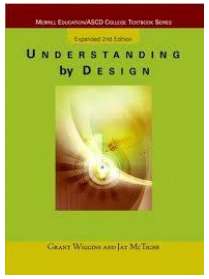
Example avenues for experiential learning:
internships, co-ops, work-study jobs,
global experiences, and original research opportunities.

What are authentic skills?

"The most important method of education ... always has consisted of that in which the pupil was urged to actual performance."—Albert Einstein.



A performance task, problem, or project, is authentic if it:



(1) *is realistic;*
(2) *requires judgment and innovation;*
(3) *engages the learner in "doing" the subject;*
(4) *replicates or simulates real-world contexts;*
(5) *assesses a learner's ability to deal with an open-ended complex task by using a repertoire of knowledge and skills;*
(6) *does not put artificial constraints on opportunities to practice, consult resources, obtain feedback, and refine performance.*

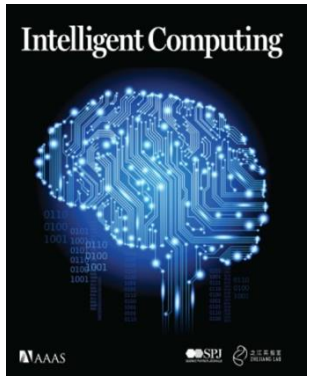
Embodied Cognition and Contextual Intelligence

EMBODIED COGNITION

What Is Missing from Contemporary AI? The World

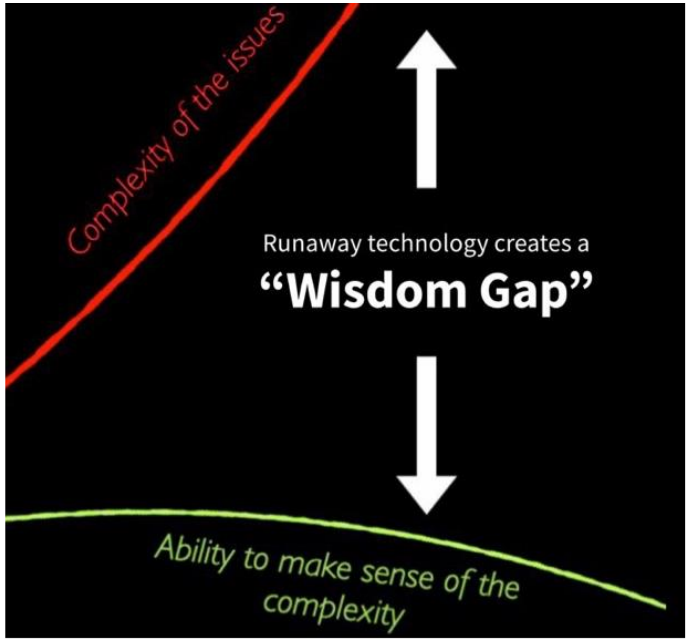
MICHAEL WOOLDRIDGE Authors Info & Affiliations

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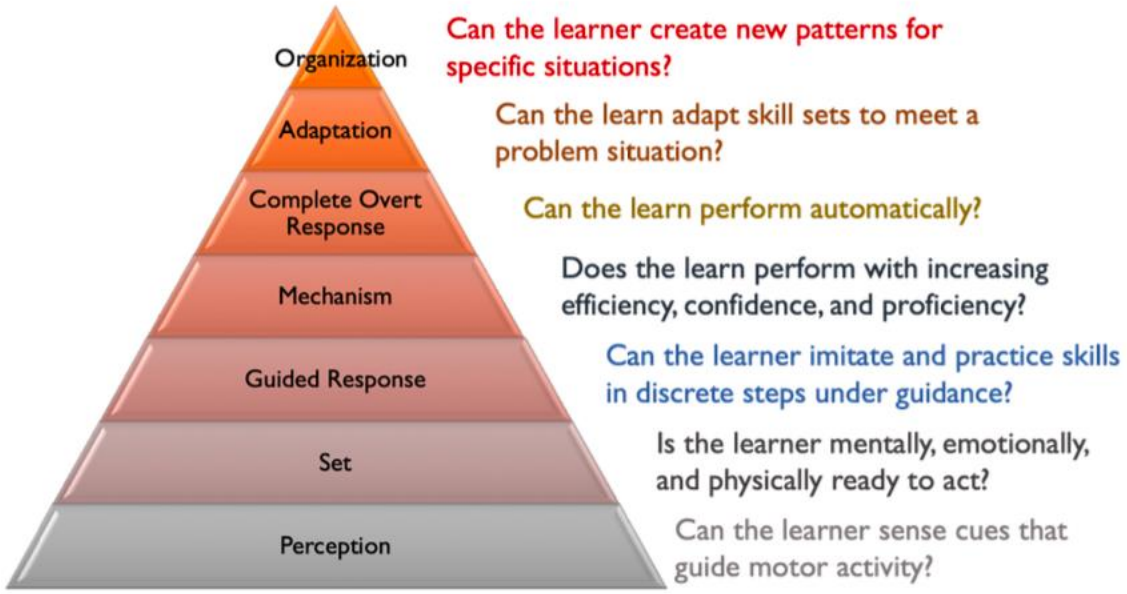
Intelligence is always embodied.

CONTEXTUAL INTELLIGENCE

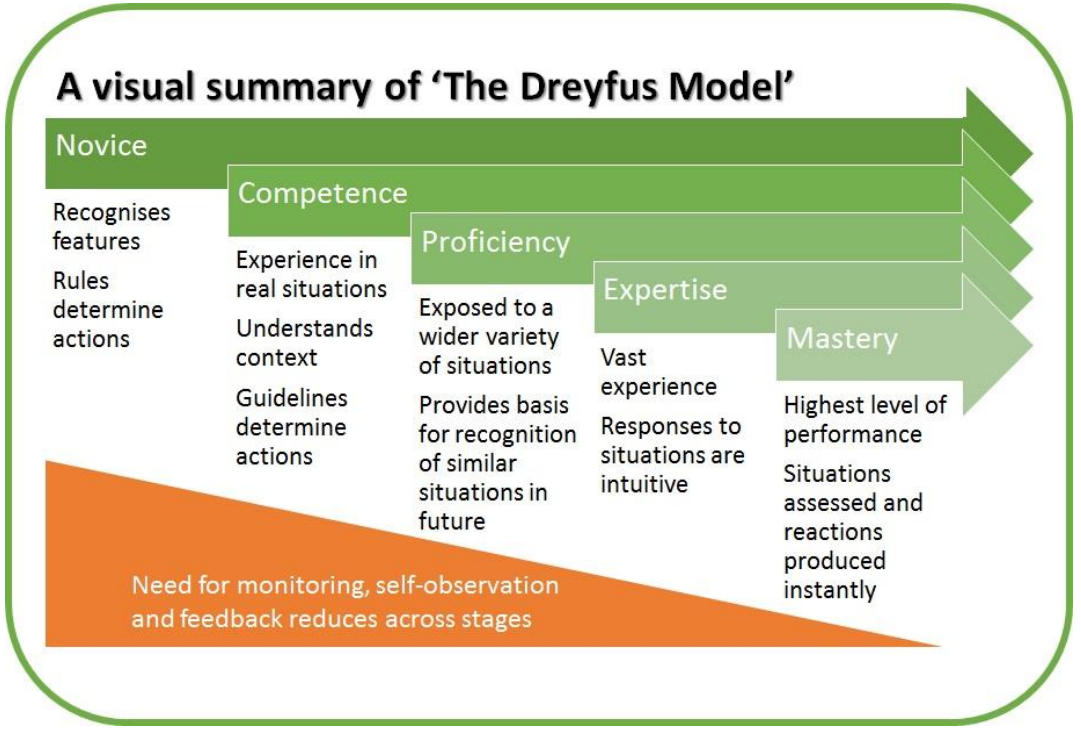


Wisdom is Contextual Intelligence

Importance of Practical Experience

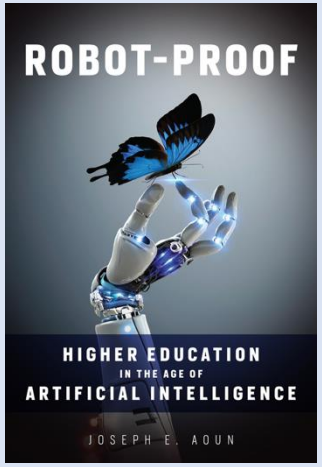


Bloom's Taxonomy for the psychomotor domain



Experiential learning is essential for developing expertise and mastery

Questions/ Discussion Mini-Break



EMBODIED COGNITION



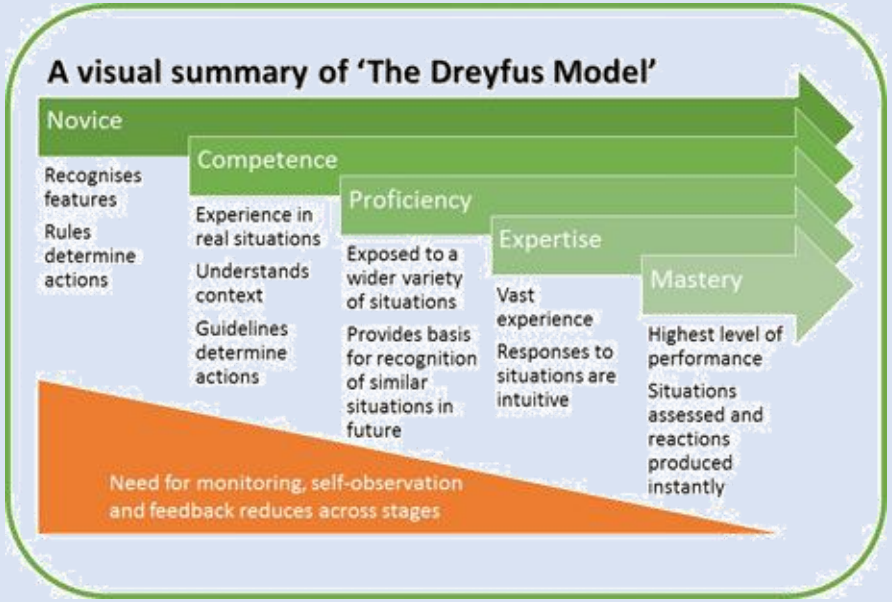


Intelligence is always embodied.

CONTEXTUAL INTELLIGENCE



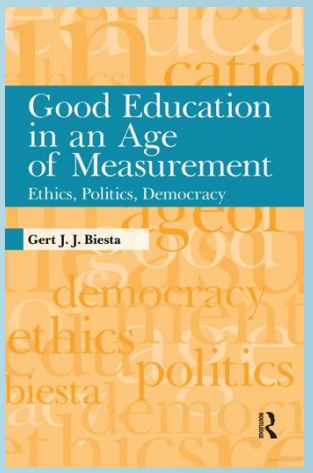
Wisdom is Contextual Intelligence



GUIDES

SELF-DEVELOPMENT

education should aim to develop free, creative, and independent individuals who can appreciate cultural achievements and contribute to society.



Apart from qualification, education also has the two important functions of "subjectification" and "socialization".

Holistic (Whole-Person) Education



THE CHARACTER IMPERATIVE



THE ECONOMIC IMPERATIVE



THE UNIQUE HUMAN POTENTIAL REALIZATION IMPERATIVE

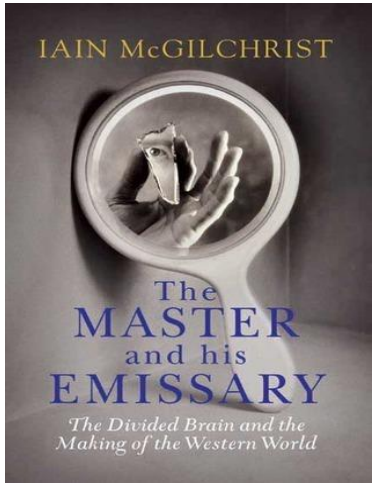


INFORMED JUDGEMENT AND CITIZENSHIP



ENGENDERING SHARED PROSPERITY

Instrumental reason and Holistic intellect



The central thesis of McGilchrist's book is about the division of the human brain into two hemispheres, each with distinct ways of interacting with the world: ***the left hemisphere*** *is adept at manipulating the world and is focused on utility, while* ***the right hemisphere*** *is associated with the perception of the broader world in all its complexity.*

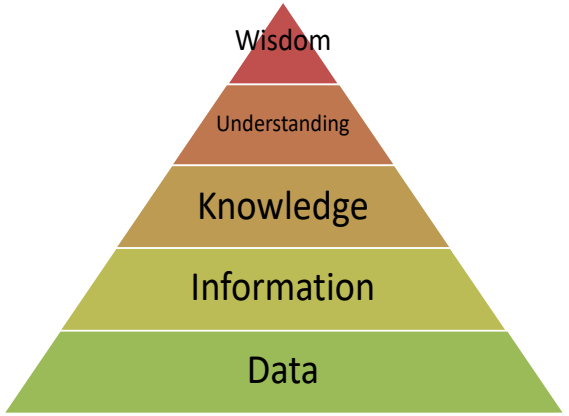
The book argues that the left hemisphere, while initially serving the right, has become overly dominant in Western society, leading to a narrow, mechanistic view of the world that misses the broader context and interconnectedness of things.



علم را بر تن زنی مارے بود
علم را بر دل زنی یارے بود

Fling it on the body, and knowledge becomes a serpent;
Fling it on the heart, and it becomes a friend.

What would educating for wisdom look like?



Wisdom emerges from an understanding of the historical, social, and cultural context. It cannot be completely programmed into logic, mathematics, models, or code

CRITICAL THINKING

—Knowing the context

—Knowing the pitfalls of rules/model/methods

—Ability to engage in informed judgement

—Knowing the limits of your knowledge

الجهل البسيط

الجهل المركّب

DUNNING-KRUGER

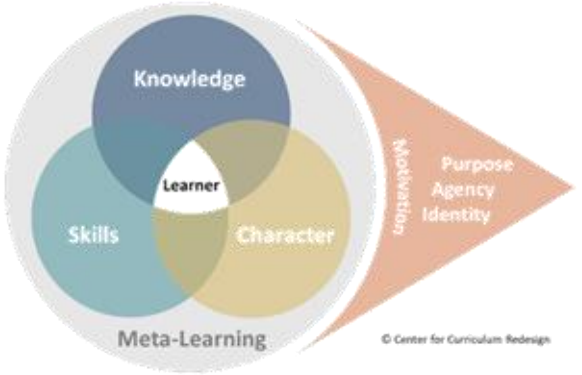
HOLISTIC THINKING

—Multidimensional/ Multidisciplinary Thinking

—Apart from Cognitive, Focus on Character, Skills, & Metacognition

—Intelligence requires both reckoning & judgement

—Dialogic Thinking

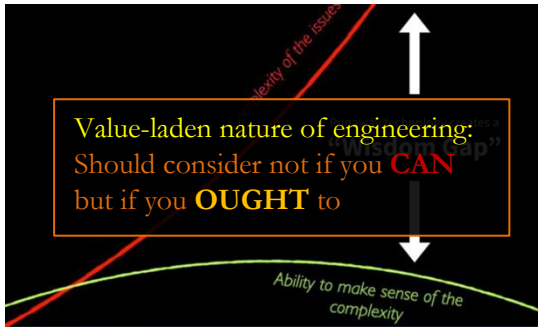


PURPOSEFUL USE OF AI

— Knowing which ends/ objectives/ purpose to pursue and prioritize

— Importance of Human Skills and Character

—Pro-Human & Pro-Social



Thank you!

For collaboration or suggestions, please reach out at:

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